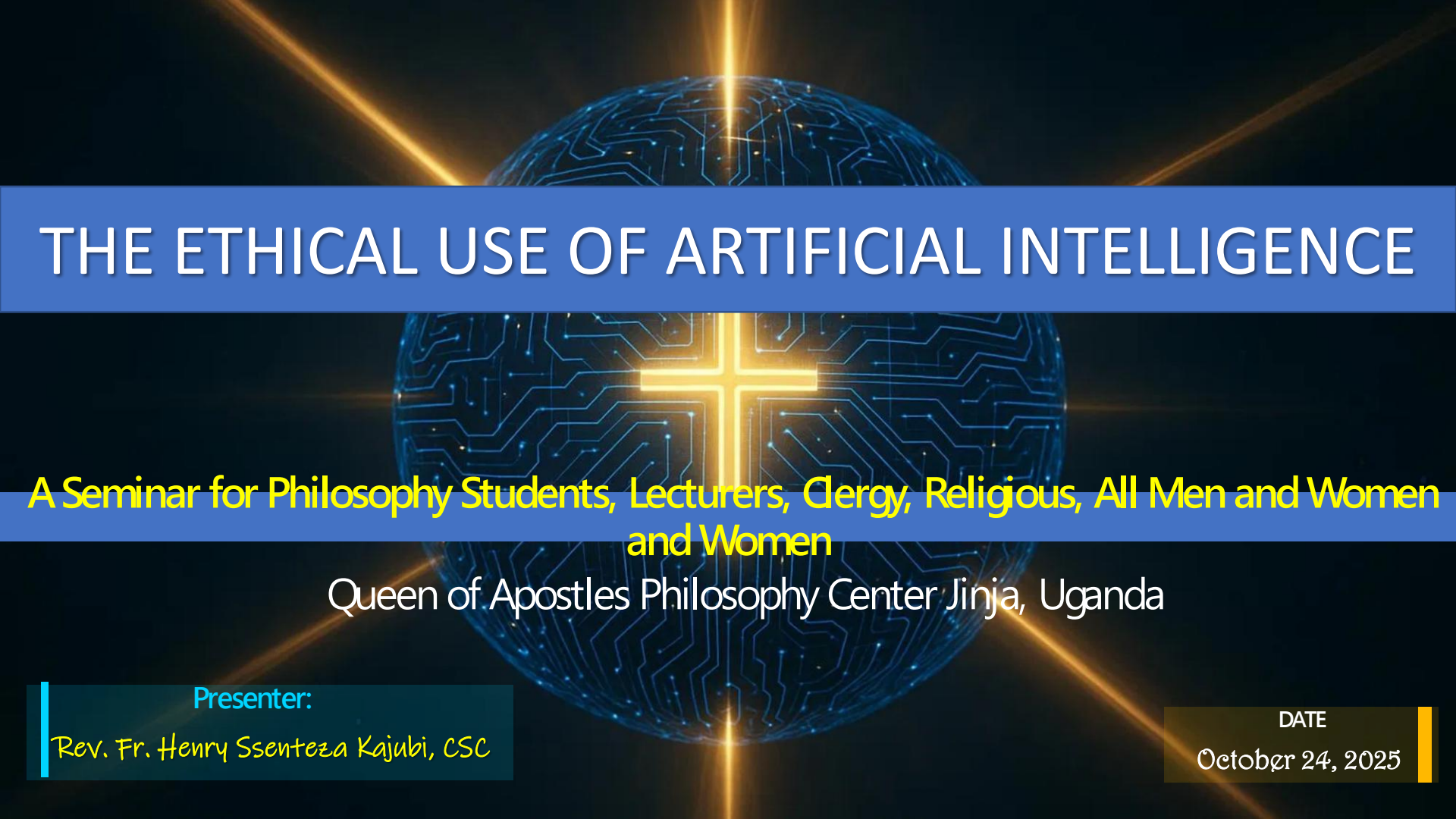




THE ETHICAL USE OF ARTIFICIAL INTELLIGENCE

A Seminar for Philosophy Students, Lecturers, Clergy, Religious, All Men and Women
and Women

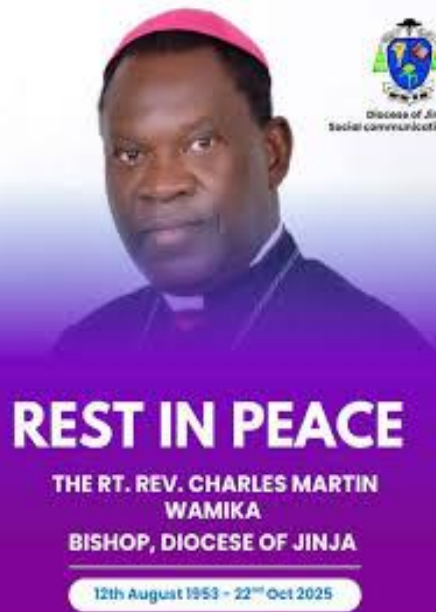
Queen of Apostles Philosophy Center Jinja, Uganda

Presenter:

Rev. Fr. Henry Ssentenza Kajubi, CSC

DATE

October 24, 2025



Lord, may Bishop Wamika's legacy inspire us to form wise, just, and compassionate leaders.
May You RIP

I Dedicate My Presentation In Memory of the Late Bishop Charles Martin Wamika, Bishop of Jinja

- 1. Formation Beyond Academics:** We remember Bishop Wamika as one who saw education as forming the whole person — intellect, spirituality, character, and mission.
- 2. A Church Unafraid of Science & Technology:** In his memory, we affirm his vision of a Church that embraces **mathematics, astronomy, and emerging technologies like AI** with confidence, not fear.
- 3. Education as Justice and Inclusion:** We honor him as a defender of seminarians in financial difficulty — a reminder that technology and AI must serve equity, not exclusion.
- 4. Personal Mentorship over Bureaucracy:** In dedication to his fatherly heart, we note that true formation — even in a digital age — must remain relational, human, and pastoral.
- 5. Preparing Leaders for the Future Church:** We commemorate his passion for forming priests ready to lead in a rapidly changing world — including AI, ethics, and responsible innovation.

In Memory



Dignity of
Human Life

Unique
Gift of
Intelligence

CREATED IN GOD'S IMAGE

"Then the LORD God formed man from the dust of the ground, and breathed into his nostrils the breath of life; and the man became a living being." – Gen 2:7

As we gather today to reflect on the ethical use of AI, we do so with wisdom both **ancient and new**.
"And he said to them, 'Therefore every scribe who has been trained for the kingdom of heaven is like the master of a household who brings out of his treasure **what is new and what is old**.'" - Mat. 13:52

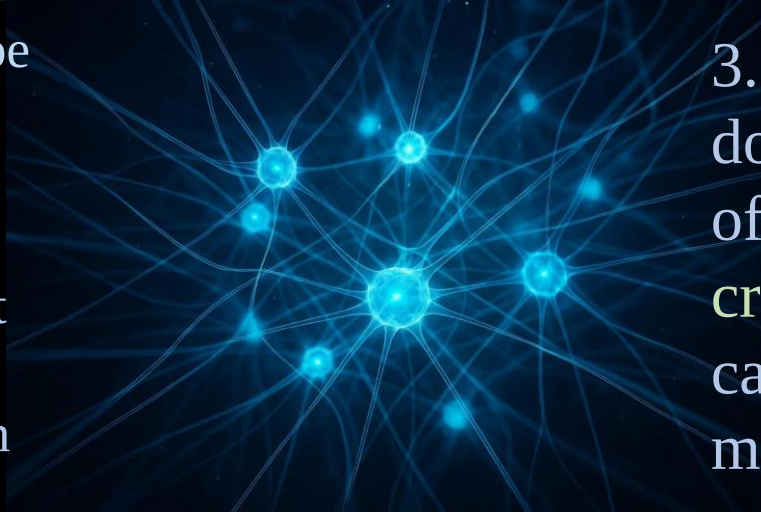
The Gift of Intelligence is an essential aspect of how humans are created in the image of God (cf. Gen 1:27). As we engage with technologies that seek to replicate human cognitive abilities, we must ground our reflection in this profound dignity.

Rapid Development of AI

Compels us to consider profound anthropological and ethical questions:

1. What does it mean to be human in an age of intelligent machines?

2. How do we ensure that technology serves the common good rather than deepening inequality?



3. What responsibilities do we bear as stewards of creation when **our** creations themselves can learn, adapt, and make decisions?

 **Pope Francis**

...has described our current era as experiencing an "epochal change," with AI standing at the heart of this transformation (Dicastery for the Doctrine of the Faith, 2025).

Foundations, Challenges, and Ethical Framework for AI

📄 Session 1: Foundations

- ✓ How AI works
- ✓ Catholic view on Technology
- ✓ Human vs AI

📄 Session 2: Ethical Challenges

- ✓ Real-world cases: bias, misinformation, creativity, employment
- ✓ Identifying risks and moral dilemmas

Session 3: Structured Response

- ✓ Open forum
- ✓ Proposing an ethical AI framework
- ✓ Grounded in CST, focused on human dignity & the common good

FOUNDATIONS OF AI ETHICS



Understanding Artificial Intelligence

- ❑ What is AI?
- ❑ How AI works?
- ❑ Machine learning?
- ❑ Neural networks?
- ❑ Rationale behind AI development



The Church's View on Technology

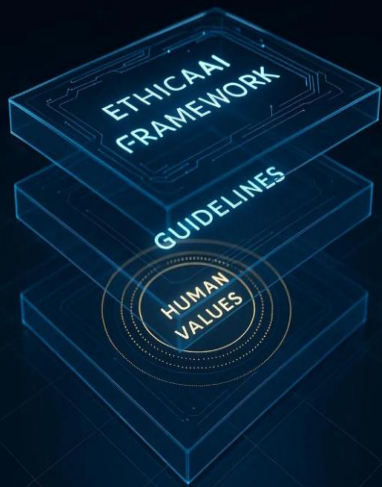
- ❖ Co-creation with God
- ❖ Stewardship
- ❖ The Common Good in CST



Human vs. Artificial Intelligence

Critical philosophical distinctions between:

1. Human Intelligence
2. Machine intelligence



What is AI?



- ✓ The term "Artificial Intelligence" was coined in 1956 by computer scientist John McCarthy, who defined it as "...making a machine behave in ways that would be called intelligent if a human were so behaving" (Stanford Encyclopedia of Philosophy, 2020).
- ✓ The Birth of AI was a formal academic field - Dartmouth Summer Research Project on AI

AI is broadly defined: **"The science and engineering of making intelligent machines, esp intelligent computer programs."** John McCarthy

AI refers to machines or software that can **perform tasks normally requiring human intelligence**, such as: Learning from experience, Problem-solving, Understanding natural language, Recognizing patterns, and Making decisions.

What is AI's Fundamental Principle?



Machines can be designed and programmed to simulate human intelligence by perceiving their environment, reasoning, learning from experience, problem-solving, and making decisions or performing tasks that typically require human cognitive abilities.

At its core, the principle is about using **computational systems to mimic or extend human-like thinking and problem-solving.**

AI is built on the principle of replicating or augmenting aspects of human intelligence in machines, enabling them to:

- 1) **Learn:** Improve performance based on data or experience (machine learning).
- 2) **Reason:** Make logical decisions or solve problems.
- 3) **Perceive:** Interpret sensory data like vision, speech, or text.
- 4) **Act:** Take actions that achieve specific goals autonomously or semi-autonomously.

HOW AI WORKS:

MACHINE LEARNING



TRAINING DATA

AI systems learn from thousands or millions of examples, analyzing patterns in vast datasets rather than following pre-programmed rules.



PATTERN RECOGNITION

Through statistical analysis, algorithms identify relationships and patterns in data, learning to recognize features and make predictions.



STATISTICAL ANALYSIS

AI uses mathematical models to find correlations, make inferences, and improve accuracy through iterative learning processes.

Example: Teaching AI to Recognize Cats

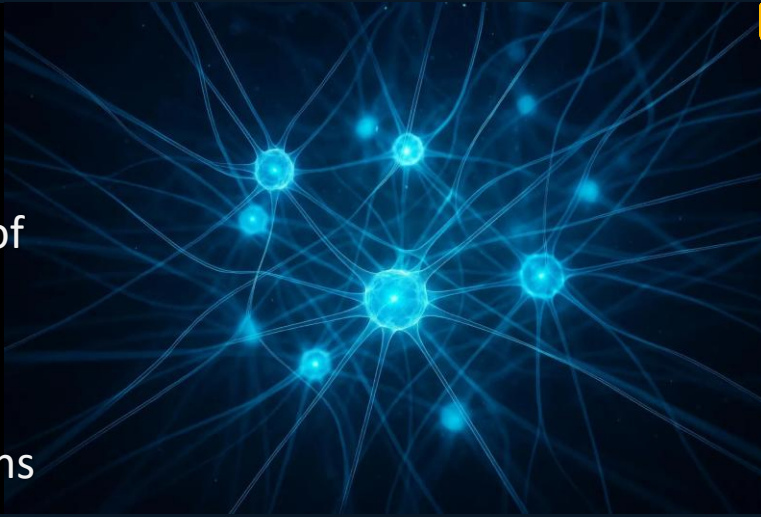
- **Collect Data:** Thousands of images labeled “cat” or “not cat.”
- **Train AI:** Neural network analyzes patterns (shapes, fur, eyes) to distinguish cats.
- **Test AI:** AI predicts cats in new, unseen images; accuracy improves with more data.
- **Application:** AI can automatically detect cats in photos/videos, e.g., smart cameras/apps

ARTIFICIAL NEURAL NETWORKS AND DEEP LEARNING

🧠 Inspired by the Brain

What Neural Networks Are

- ✓ Inspired by the human brain — made of layers of artificial “neurons”
- ✓ Each neuron processes input, passes results forward to make decisions



📈 Deep Learning Process

What Deep Learning Does

- ✓ Adds many hidden layers → learns complex patterns automatically
- ✓ Powers AI tasks like image recognition, translation, voice assistants, etc

Deep learning models have *many layers*, and each layer learns a more *advanced level of understanding* than the previous one.

- ❖ Early layers detect simple patterns (like edges or colors).
 - ❖ Deeper layers combine those into meaningful features (like eyes, faces, objects).
- **That is why it's called “deep”**

The data is processed step by step through many layers of understanding, from simple to complex.



THE RATIONALE BEHIND AI DEVELOPMENT



Augmenting Human Capabilities

AI frees us from tedious tasks, extends our cognitive reach, and enables us to process information at unprecedented scales



Solving Complex Problems

Healthcare diagnosis, climate modeling, poverty alleviation, education personalization—challenges requiring vast data analysis



Accelerating Discovery

AI processes information at speeds impossible for humans, humans, potentially accelerating scientific breakthroughs and innovation

"AI can be trained on the results of human creativity and then generate new artifacts with a level of speed and skill that often rivals or surpasses what humans can do." *Vatican, Rome Call for AI Ethics, 2020 (Principle on Algorithmic Responsibility) - Microsoft, IBM, and the UN's FAO, emphasizing responsible and ethical AI*



TECHNOLOGY AS CO-CREATION

"When men and women provide for themselves and their families in such a way as to be of service to the community as well, they can rightly look upon their work as a prolongation of the work of the creator."

— **Gaudium et Spes, 34**

- ✓ GS views scientific and technological endeavor as a participation in God's creative activity

Biblical Foundation

(Genesis 2:15)

*"The Lord God took the man and put him in the garden of **Eden to till it and keep it.**"*

"The Church encourages the advancement of science, technology, the arts, and other forms of human endeavor, viewing them as part of the collaboration of man and woman with God." — Antiqua et Nova, 2025

Session 1



"The LORD God placed the man in the Garden of Eden to till it and keep it." Gen 2:15

⚠ WARNING **The Technocratic Paradigm**
A dangerous belief that technology can solve all problems without ethical guidance or moral discernment

Pope Francis – Technology is not neutral, it embodies the values and priorities of those who create and deploy it

STEWARDSHIP: To Till & Keep

🌱 **ABAD: To Till, to serve, to work, to cultivate, to develop**

We are called to use our Intelligence and Creativity to develop technology that serves humanity

🛡 **SHAMAR: To keep, to guard, to preserve**

We are called to protect creation, ensuring that our Technologies respect human dignity and the environment

A CALL TO TECHNOLOGICAL DISCERNMENT

Technology, rightly understood, is an expression of this dual vocation: to develop AI that serves humanity while protecting human dignity and the common good

"The indisputable benefit that humanity can draw from technological progress depends on the degree to which new possibilities are placed at the service of human dignity and the common good." — **Laudato Si', 102**



HUMAN VS. ARTIFICIAL INTELLIGENCE



HUMAN

1. **Personhood** – tied to consciousness and soul
2. **Intuition** – direct grasp of truth
3. **Creativity** – genuine originality
4. **Moral Reasoning** – ethical judgment
5. **Emotional Depth** – authentic feeling
6. **Spiritual Awareness** – transcendence



ARTIFICIAL

1. **Functionality only** – no consciousness
2. **Pattern recognition** – statistical inference
3. **Recombination** – not genuine creation
4. **No moral agency** – follows programming
5. **No emotions** – simulates responses
6. **No spiritual capacity** – material only

"In the case of humans, intelligence is a faculty that pertains to the person in his or her entirety, her entirety, whereas in the context of AI, Intelligence is understood as functionality

— Antiqua et Nova, Vatican Document, 2025

INTELLECTUS ET RATIO

St. Thomas Aquinas distinguished two facets of human intelligence: intelligence:



INTELLECTUS

The **intuitive grasp of truth** — the ability to apprehend reality directly with the "eyes" of the mind. The capacity for wisdom, for seeing the whole, for understanding first principles.

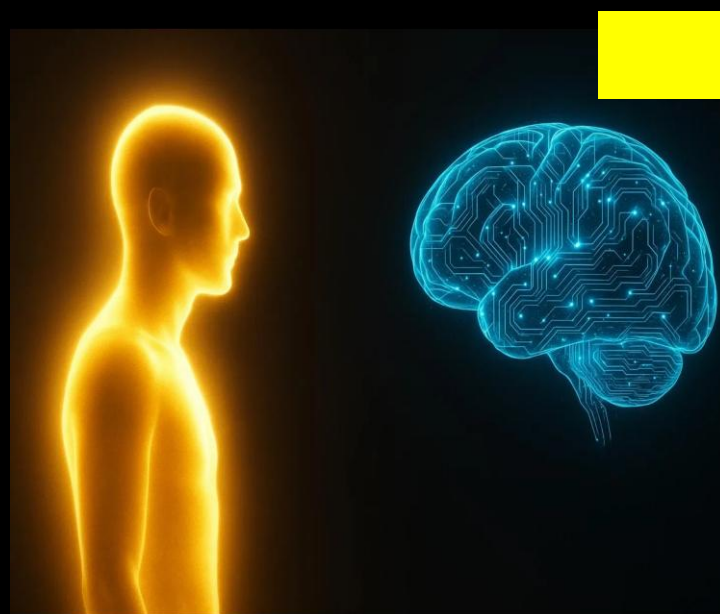
→ **AI has NO intellectus**



RATIO

Discursive reasoning — the step-by-step analytical process that leads from premises to conclusions. Logical deduction, pattern recognition, calculation.

→ **AI has limited, mechanistic ratio only**



AI's advanced features give it sophisticated abilities to perform tasks, but **not the ability to think**

- Antiqua et Nova, 2025

PHILOSOPHICAL FRAMEWORKS

1. KANTIAN ETHICS: Human Dignity



Rational beings possess inherent worth and must always be always be treated as ends in themselves, never merely as as means. AI must respect human autonomy and not to reduce reduce individuals to data pints.

Laws governing AI must be universally acceptable and beneficial beneficial to all



2. VIRTUE ETHICS: Human Flourishing

Focuses on character and cultivation of virtues: prudence, justice, fortitude, temperance.

What kind of people do we become when we create create and use AI?

AI should help us cultivate wisdom and compassion, not compassion, not diminish our moral capacities.

3. CATHOLIC SOCIAL TEACHING

Seven principles provide comprehensive framework: dignity of the human person, community, rights and responsibilities, option for the poor, dignity of work, solidarity, care for creation.

Evaluate AI not by efficiency or profit, but by impact on human human dignity, justice, and the common good.



SESSION 1 SUMMARY

FOUNDATIONS OF AI ETHICS



HOW AI WORKS

WORKS

- ✓ Machine learning from training data
- ✓ Neural networks data with layered architecture
- ✓ Pattern recognition and statistical analysis
- ✓ Augments human capabilities



CHURCH'S VIEW

- ✓ Technology as co-creation with God
- ✓ Biblical mandate: till and keep
- ✓ Warning against technocratic paradigm
- ✓ Must serve common good



HUMAN VS AI

- ✓ Human: personhood soul, consciousness,
- ✓ AI: functional only, no thinking
- ✓ Intellectus vs limited ratio
- ✓ Three ethical frameworks

Session 1

CONTEMPORARY ETHICAL CHALLENGES



Examining real-world AI ethics issues through case studies

⚠️ **Challenge 1: Algorithmic Bias and Discrimination**

How AI systems perpetuate and amplify human biases

🧠 **Challenge 2: Deepfakes and the Crisis of Truth**

AI-generated misinformation threatening democracy and trust

🎮 **Challenge 3: AI and Human Creativity**

Questions about authorship, originality, and human uniqueness

🤖 **Challenge 4: The Dignity of Work**

AI's impact on employment and human fulfillment



CHALLENGE 1

ALGORITHMIC BIAS AND DISCRIMINATION

❗ The Core Problem

AI systems do not create bias from nothing—they amplify the biases in training data and societies that produce them

☰ How Bias Enters AI

- Training data reflects historical discrimination
- Algorithm design by non-diverse teams
- Deployment contexts without equity testing

🦋 FACIAL RECOGNITION BIAS

Research by Buolamwini and Gebru found facial recognition systems errors (34%) for darker-skinned women compared to lighter skinned men

Result: Wrongful arrests, surveillance targeting minorities

💼 HIRING ALGORITHM BIAS

Amazon's AI recruiting tool systematically downgraded Women's resumes because it was trained on hired male dominated data

Result: Perpetuation of gender discrimination



DATA ANALYSIS

CASE STUDIES: BIAS IN ACTION



FACIAL RECOGNITION: Robert Williams Wrongful Arrest (2020)

African American man wrongfully arrested in Detroit after facial recognition system misidentified him. Research by Buolamwini & Gebru found error rates up to **34% higher** for darker-skinned women compared to lighter-skinned men. AI trained predominantly on white faces fails to accurately recognize people of color.



HIRING DISCRIMINATION: Amazon Recruiting Tool (2018)

Amazon's AI recruiting tool systematically **downgraded women's résumés** because it was trained on historical hiring data from a male-dominated tech industry. The algorithm penalized résumés containing words like "women's" (e.g., "women's chess club captain"). Amazon ultimately scrapped the system.



HEALTHCARE DISPARITIES: Risk Prediction Algorithm (Obermeyer Study)

Healthcare algorithm used to guide treatment decisions showed significant racial bias. Black patients were assigned **lower risk scores** than equally sick white patients because the algorithm used healthcare spending as a proxy for health needs—but Black patients historically have less access to care and lower spending.



CHALLENGE 2

DEEPAKES AND THE CRISIS OF TRUTH

"You will know the truth, and the truth will make you free."— **John 8:32, NRSV**

What Are Deepfakes?

AI-generated synthetic media that can create hyper-realistic fake videos, images and audio of people saying or doing things they never did

How the Technology Works

- The technology uses deep learning algorithms to analyze and manipulate visual and audio content.
- Neural networks are trained on thousands of images and videos to learn facial movements, voice patterns, and expressions, then **generate convincing fabrications**.

DEEPAKES: THREATS TO SOCIETY



1. POLITICAL MANIPULATION

Deepfake videos of political leaders making false statements, declarations of war, or inflammatory remarks can **destabilize democracies**, incite violence, and manipulate election outcomes. The technology enables unprecedented levels of political disinformation that threaten the integrity of democratic processes and public discourse.



2. EROSION OF TRUST AND THE "LIAR'S DIVIDEND"

When any video can potentially be fake, **authentic evidence loses credibility**. This creates the "liar's dividend"—bad actors can dismiss genuine evidence of wrongdoing by claiming it's a deepfake. The result is a collapse of shared reality, where journalism, legal evidence, and eyewitness testimony all become suspect.



3. PERSONAL HARM AND DIGNITY VIOLATIONS

Deepfake technology is weaponized against individuals through **non-consensual pornography**, identity theft, financial fraud, and character assassination. Victims—disproportionately women—face profound violations of dignity, reputation damage, and psychological trauma with limited legal recourse.

The crisis of truth threatens the very foundation of democratic society, journalism, and human truth as sacred value.

We must defend relationships grounded in the Biblical promise

AI AND HUMAN CREATIVITY

CHALLENGE 3

❓ Can AI Be Truly Creative?

AI can generate impressive artifacts—paintings, music, poetry. But does it possess **genuine creativity** or merely sophisticated **recombination** of existing patterns?

"AI can be trained on the results of human creativity and creativity and then generate new 'artifacts' with a level of level of speed and skill that often rivals or surpasses what surpasses what humans can do." — Antiqua et Nova, 2025

The Philosophical Question

💡 True creativity involves **originality**, **intentionality**, and **meaning-making** rooted in human experience, emotion, and consciousness—**qualities AI lacks**.

🎨 Examples of AI-Generated Art

"The Next Rembrandt" (2016)

AI analyzed Rembrandt's paintings and generated a new "Rembrandt" portrait. Technically impressive, but did the AI **understand** Rembrandt's vision of human dignity? Did it **intend** to convey meaning?

Schubert's Unfinished Symphony

AI completed Schubert's Symphony No. 8 by analyzing his compositional patterns. Beautiful result, but lacking the **human soul** and **lived experience** that informed Schubert's work.

❓ Questions of **Authorship, originality**, and what

makes humans **creative** remain unresolved.



CHALLENGE 4

THE DIGNITY OF WORK

✝ Catholic Teaching on Work

Work is not merely an economic transaction—it is a fundamental dimension of **human existence**.

A participation in God's ongoing creation.

"Through work man not only transforms nature, adapting it to his own needs, but he also achieves fulfillment as a human being."

— Laborem Exercens, Pope John Paul II, 1981

- ✓ Work as human fulfillment and self-realization
- ✓ Work as participation in God's creation
- ✓ Work as contribution to the common good

🤖 AI's Impact on Employment

Automation threatens to displace millions of workers, raising urgent questions about **human dignity** and the meaning of work in an AI-driven economy.

⚠ THE CHALLENGE

- Job displacement and unemployment
- Loss of purpose and identity
- Growing economic inequality
- Erosion of worker solidarity

💡 AI must not reduce workers to mere economic units. Technology should enhance human work, not replace human workers entirely.

WORK AND HUMAN DIGNITY



"Work is a good thing for man—a good thing for his humanity—because through work man not only transforms nature, adapting it to his own needs, but he also achieves fulfillment as a human being."

— *Laborem Exercens*, Pope John Paul II, 1981

👤 Work as Human Fulfillment

Catholic teaching views work not merely as economic necessity, but as a fundamental dimension of **human existence** through which we participate in God's creative activity and develop our personhood.

🤖 The Challenge of Automation

AI-driven automation threatens to displace millions of workers. The ethical question is not whether AI **can** replace human labor, but whether it **should** and how we ensure workers retain dignity and meaningful participation.

🤝 Solidarity Demands

We must ensure that the benefits of AI are **shared broadly**, not concentrated Among tech elites. This requires retraining programs, social safety nets, and Economic policies that prioritize human dignity over profit maximization.

AI must be deployed in ways that enhance human work, not eliminate human workers, technology should serve common good, not just corporate efficiency



SESSION 2 SUMMARY

CONTEMPORARY ETHICAL CHALLENGES



ALGORITHMIC BIAS

- ✓ Perpetuates discrimination
- ✓ Facial recognition errors
- ✓ Hiring discrimination
- ✓ Healthcare disparities



CRISIS OF TRUTH

- ✓ Deepfake misinformation
- ✓ Political manipulation
- ✓ Erosion of trust
- ✓ Threat to democracy



AI CREATIVITY

- ✓ Recombination vs creation
- ✓ Questions of authorship
- ✓ Lack of intentionality
- ✓ Human uniqueness



DIGNITY OF WORK

- ✓ Job displacement threat
- ✓ Work as fulfillment
- ✓ Economic inequality
- ✓ Solidarity demands

ETHICAL FRAMEWORK FOR AI



Practical Ethical Framework Grounded in CST



Human Dignity First

AI must serve humanity, never reduce persons to data



Transparency and Justice

Explainable AI decisions and fairness across all populations



The Common Good

Benefits for all, especially the poor and vulnerable



Catholic Social Teaching

Seven principles guiding ethical AI development and use



Education and Action

AI literacy, eliminating bias, and responsible use

HUMAN DIGNITY FIRST



**AI must serve humanity,
not vice versa.**

Every human being possesses inherent dignity as one created in the image of God. Technology must respect and enhance this dignity, never

Respect Human Autonomy

AI systems must respect human freedom and decision-making capacity

Meaningful Human Control

Humans must retain ultimate authority over critical decisions

Never Reduce Persons to Data

Individuals are not mere data points or objects of manipulation

**"So God created humankind in his image, in the
image of God he created them" – Gen 1:27**

This dignity is **inviolable**

+ and must guide all AI development.

TRANSPARENCY AND EXPLAINABILITY



Catholic teaching on **truth** and **accountability** demands that AI systems be open to scrutiny, especially when they affect human dignity and rights.

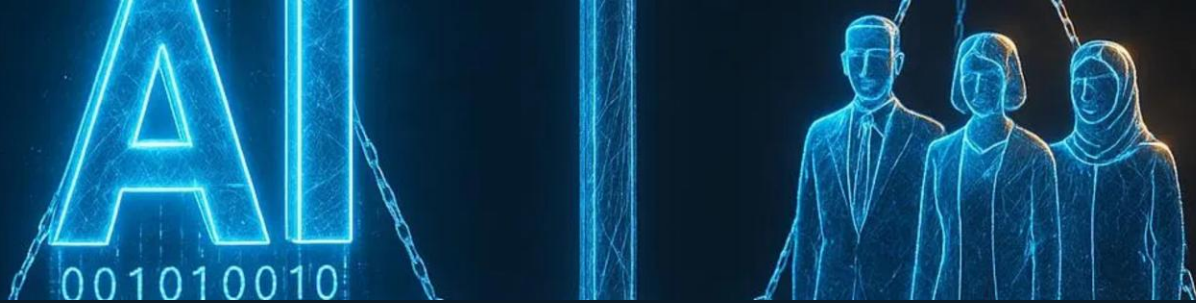
AI decision-making must be understandable and accountable
No “BLACK-BOX” systems for decisions affecting human lives.

💡 Why Transparency Matters

- ✓ **Accountability:** We must be able to identify and correct errors
- ✓ **Trust:** People deserve to know how decisions are made
- ✓ **Human Oversight:** Meaningful control requires understanding
- ✓ **Justice:** Those affected have right to explanation

📋 REQUIREMENTS FOR TRANSPARENT AI

- Explainable algorithms that show reasoning process
- Audit trails documenting data sources and decisions
- Human review mechanisms for high-stakes decisions
- Clear disclosure when AI is making decisions



ETHICAL PRINCIPLE 3

JUSTICE AND FAIRNESS

AI must not perpetuate discrimination. Systems must be tested across diverse populations and bias must be addressed **proactively**, not reactively.

Why Justice Matters in AI

AI systems trained on historical data often **perpetuate and amplify** existing societal biases against marginalized groups—racial minorities, women, the poor, and persons with disabilities.

Catholic Social Teaching demands **preferential option for the poor** and protection of the vulnerable. AI must advance justice, not undermine it.

"Learn to do good; seek justice, rescue the oppressed, defend the orphan, plead for the widow."

— Isaiah 1:17. NRSV

Session 3

REQUIREMENTS FOR FAIR AI



1. RIGOROUS TESTING ACROSS DIVERSE POPULATIONS

Test AI systems on diverse demographic groups before deployment. Measure performance disparities across race, gender, age, disability status, and socioeconomic background. Address inequities before harm occurs.



2. DIVERSE DEVELOPMENT TEAMS

Include perspectives from marginalized communities in AI development. Diverse teams help identify blind spots, challenge assumptions, and ensure systems serve all people, not just privileged groups.



3. PROACTIVE BIAS DETECTION

Identify and correct bias before harm occurs. Use bias detection tools, conduct algorithmic audits, and implement fairness metrics throughout the development lifecycle.



4. CONTINUOUS MONITORING OF OUTCOMES

Track outcomes across demographic groups over time. Bias can emerge or worsen after deployment as systems interact with systems interact with real-world data. Ongoing monitoring is essential.



5. LEGAL ACCOUNTABILITY AND ENFORCEMENT

Enforce anti-discrimination laws in AI systems. Establish clear legal liability for biased AI, create regulatory frameworks, and ensure victims have recourse when harmed by discriminatory algorithms.



Justice demands that AI systems be **fair by design**, not as an afterthought. We must build equity into every Stage of AI development and deployment.

ETHICAL PRINCIPLE 4

THE COMMON GOOD

AI should benefit **all people**, especially the **poor and vulnerable**.
Technology must serve the common good, not just private profit.

✚ Preferential Option for the Poor

Catholic Social Teaching demands that we evaluate technology by how it affects the most vulnerable. AI development must prioritize the needs of marginalized communities, not just wealthy consumers.

🌐 Key Requirements

- ✓ Address the Digital Divide Ensure equitable access to AI benefits
- ✓ Promote Global Solidarity Share AI benefits across borders
- ✓ Prioritize Human Needs Healthcare, education over luxury applications
- ✓ Prevent Concentration of Power Distribute AI capabilities widely

👉 "We are one human family" (Fratelli Tutti). AI must strengthen our **solidarity** and promote the **flourishing of all**, not deepen inequality





ETHICAL FRAMEWORK

CATHOLIC SOCIAL TEACHING: SEVEN PRINCIPLES



1. HUMAN DIGNITY

Every person is sacred, made in God's image. AI must respect this inherent worth.



2. COMMUNITY

We are social beings. AI should strengthen human connection, not isolate.



3. RIGHTS

Human rights come with duties. AI developers bear responsibility for impact.



4. OPTION FOR POOR

Prioritize needs of vulnerable. AI benefits must reach marginalized first.



5. DIGNITY OF WORK

Work is human fulfillment. AI should enhance work, not eliminate workers.



6. SOLIDARITY

We are one human family. AI must promote global unity, not division.



7. CREATION CARE

Stewardship of Earth. AI's environmental impact must be sustainable.

†These seven principles provide a **comprehensive moral framework** for evaluating and guiding AI development and deployment.

SOLIDARITY AND GLOBAL CONNECTION



🌐 "We are one human family"

"Digital connectivity is not enough to build bridges. It is not capable of uniting humanity... The digital world is a public square, a meeting-place where we can either encourage or demean one another, engage in a meaningful discussion or unfair attacks." — Fratelli Tutti, Pope Francis, 2020

🔗 AI Must Promote Unity, Not Division

Global Cooperation: AI development requires international collaboration and shared ethical standards

Address Digital Divide: Ensure AI benefits reach developing nations and marginalized communities

Prevent AI Nationalism: Avoid weaponization of AI or using technology to dominate others

Strengthen Human Bonds: Use AI to facilitate genuine human connection, not replace it

Solidarity means recognizing that we are interdependent . AI must be developed and deployed in ways that strengthen our common humanity Humanity and promote the flourishing of all peoples

THE IMPERATIVE OF AI EDUCATION



We must **learn how AI works**, identify its biases, use it responsibly. AI literacy is no longer optional—it is a **moral imperative**

🎓 Five Components of AI Literacy

1. Understanding How AI Works

Grasp technical basics: machine learning, training data, algorithms

2. Identifying Bias and Errors

Recognize when AI produces discriminatory or inaccurate results

3. Evaluating AI Outputs Critically

Question AI-generated content, verify sources, maintain skepticism

4. Using AI Responsibly and Ethically

Apply ethical principles, respect privacy, avoid harmful applications

5. Understanding Societal Implications

Grasp AI's impact on work, democracy, justice, and human dignity

💡 The Church, universities, and communities must **prioritize AI education** to empower people to engage technology with **wisdom and virtue**

LEARNING TO ELIMINATE ERRORS AND BIASES



Eliminating bias requires **intentional effort** **technical expertise**, and ongoing commitment. We must learn how to identify and correct AI errors proactively.

✍ 1. RIGOROUS TESTING ACROSS DIVERSE POPULATIONS

Test AI systems on diverse demographic groups before deployment. Measure performance disparities and address them before harm occurs.

📊 2. CONTINUOUS MONITORING OF OUTCOMES

Track AI decisions and outcomes over time across different populations. Bias can emerge or worsen after deployment.

👥 3. DIVERSE DEVELOPMENT TEAMS

Include people from marginalized communities in AI development. Diverse perspectives help identify blind spots and biases.

🔍 4. ETHICAL REVIEW BOARDS AND AUDITS

Establish independent oversight mechanisms to review AI systems for bias, fairness, and ethical compliance.

🎓 **AI literacy** is essential for everyone—developers, users, policymakers, and citizens. We must all learn to identify and challenge biased systems.



USER RESPONSIBILITY

AI MORALITY DEPENDS ON HUMAN CHOICES

AI is a **tool**. Its moral character depends on the **values, intentions, and choices** of those who create and use it.

GOOD USES

💖 Healthcare

Diagnosis, treatment planning, drug discovery

🎓 Education

Personalized learning, accessibility tools

🌐 Accessibility

Assistive technologies for persons with disabilities

🔬 Scientific Research

Climate modeling, disease prevention, innovation

❌ BAD USES

📺 Mass Surveillance

Violating privacy, tracking citizens without consent

💣 Autonomous Weapons

Lethal AI systems that kill without human judgment

🗣️ Manipulation

Disinformation, deepfakes, psychological manipulation

⚖️ Discrimination

Biased algorithms perpetuating injustice

Session 3

We must cultivate **virtues**—wisdom, justice, compassion—to guide our use of AI toward the **common good**



CALL TO ACTION

THE CHURCH'S ROLE IN AI ETHICS



1. EDUCATION

Teach AI ethics in Catholic schools, universities, and seminaries. Equip the faithful with knowledge to engage technology wisely.



2. DIALOGUE

Facilitate conversations between technologists, ethicists, theologians, and policymakers to bridge divides.



3. ADVOCACY

Promote just AI policies and regulations that protect human dignity, especially for the poor and vulnerable.



4. MODELING ETHICAL PRACTICE

Demonstrate responsible AI use in Church institutions, showing how technology can serve the common good.



5. SPIRITUAL DISCERNMENT

Help people navigate AI's impact on human meaning, purpose, and relationships through prayer, reflection, and pastoral care. Guide the faithful to use technology in ways that deepen rather than diminish their humanity.

CONCLUSION: FAITH, HOPE, AND LOVE

"For the whole law is summed up in a single commandment, 'You shall love your neighbor as yourself.'" — Galatians 5:14, NRSV

+ FAITH

Trust in the inherent **dignity of every human person** created in God's image. Technology must serve this sacred truth.

🐦 HOPE

Confidence that AI, when guided by **wisdom and virtue**, can serve the common good and advance human flourishing.

♥ LOVE

Commitment to use AI for our **neighbor's flourishing**, especially the poor and vulnerable. Love must guide every decision.



AI must be grounded in **human dignity**, **justice**, and **love of neighbor**. This is our calling.

Session 3

Thank you - May God Bless you